Financing Population Aging:

Tax Mix, Public Debt, and Informality in Colombia

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Abstract

Population aging raises social-security spending even when the real benefit or service received by each older adult remains constant. This paper asks how that permanent fiscal pressure should be distributed across consumption, formal payroll, and capital-income taxes in an economy with a large informal sector. I build a stationary general-equilibrium model with formal capital, labor-intensive informal production, imperfect substitution between formal and informal consumption goods, endogenous labor informality, public debt, and convex tax-collection costs. DANE projections raise the ratio of people aged 65 and over to the population aged 15–64 from 15.02% in 2024 to 27.28% in 2050 and 48.17% in 2070. At Colombia's statutory debt anchor of 55 percent of GDP, the model assigns 82.5% of the 2050 revenue requirement to consumption and 17.5% to payroll; the stationary capital-income tax is zero. The associated effective rates are 17.4% and 4.5%, and informality is 56.0%. As aging intensifies, the payroll share rises because concentrating revenue in a single base has a convex resource cost. Without that collection friction, the model returns the corner solution of 100.0% consumption taxation. Raising the debt anchor does not finance aging in the long run: it increases the stationary interest bill and required revenue. These numerical shares are mechanism results, not a tax proposal. The benchmark has one representative household and provisional collection costs; it therefore excludes distributional incidence, intragenerational consumption variance, and a perfect-foresight transition.

Keywords: population aging; optimal tax mix; public debt; informality; Colombia. **JEL:** E62; H21; H55; H63; J46.

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1 Introduction

Population aging changes the fiscal problem even if social-security rules do not change. Suppose that every older adult receives a fixed real amount of pensions, health services, or long-term care. The government's outlay per working-age adult then rises with the ratio of older people to the population that supplies most labor and pays most taxes. Debt can postpone part of the adjustment. It cannot replace the permanent revenue that a permanently older population requires.

This paper asks which combination of consumption, formal-payroll, and capital-income taxes finances that pressure at the lowest stationary welfare cost. The question is particularly relevant for Colombia because the tax bases are not equally broad. A payroll tax applies directly to formal employment. A consumption tax reaches formal purchases and only part of informal transactions. A capital-income tax falls on the input used by the formal sector but not by the labor-intensive informal sector. Each instrument therefore changes both the revenue collected from its own base and the allocation of labor and capital between sectors.

The demographic input comes directly from DANE's 2025 national population projections. Let N_t^o denote the population aged 65 and over and N_t^y the population aged 15–64. DANE projects that

$$d_t = \frac{N_t^o}{N_t^y}$$

rises from 15.02% in 2024 to 27.28% in 2050 and 48.17% in 2070 (Departamento Administrativo Nacional de Estadística, 2025b). If the real outlay per older adult is $\bar{g}_o$, social-security spending per working-age adult is the identity

$$G_t^{SS} = \bar{g}_o d_t.$$

This formulation turns population aging into a fiscal requirement without assuming an arbitrary expenditure shock.

The model adds four margins to that identity. First, formal and informal consumption goods enter a constant-elasticity-of-substitution aggregator. They are substitutes, but not perfect substitutes. Second, only formal production uses capital. Third, workers choose formal or informal employment according to after-tax relative wages and heterogeneous sector preferences. Fourth, raising a tax has a convex real collection cost that represents administration, compliance, and resources lost through avoidance and enforcement. This last margin matters for the tax mix: without it, spreading revenue across bases has no direct benefit.

Public debt enters through the government's flow budget and a fiscal reaction rule. The benchmark uses the 55-percent-of-GDP net-debt anchor established by Law 2155 of 2021; the same law defines a 71-percent limit (República de Colombia, 2021). The model varies the anchor between 40 and 70 percent to show how $\bar{q}$ changes the stationary interest bill. This is a long-run comparative static. It does not quantify the possible gain from issuing debt during the demographic transition and repaying it later.

The calibration matches 55.4% labor informality and normalizes initial public pension spending to 5.7% of GDP, the value reported for Colombia by the OECD (Departamento Administrativo Nacional de Estadística, 2025a; Organisation for Economic Co-operation and Development, 2023). The remaining parameters, including the collection-cost coefficient and the fraction of social-security spending that purchases real goods and services, are provisional. The numerical exercise should therefore be read as a disciplined mechanism comparison.

Three results organize the paper. First, the frictionless benchmark produces a corner: all revenue is raised from consumption. A CES distinction between formal and informal goods does not by itself generate an interior tax mix. Second, common convex collection costs produce a combination of consumption and payroll taxes. At the 55-percent debt anchor, the consumption share falls from 92.5% in 2024 to 82.5% in 2050 and 75.0% in 2070. The payroll share rises correspondingly. The stationary capital-income tax remains zero because this representative-household model contains neither a distributional motive nor capital rents. Third, a higher debt anchor raises long-run required revenue. Moving the anchor from 40 to 70 percent of GDP lowers stationary composite consumption by about 0.6 percent in the 2050 demographic equilibrium.

The next section relates the exercise to the literature. Section 3 presents the model. Section 4 describes the calibration and the numerical policy experiment. Section 5 reports the results. Section 6 states the limitations that restrict their policy interpretation. The appendix describes the algorithm.

2 Related literature

The paper connects three bodies of work. The first is the optimal-tax literature. Diamond and Mirrlees (1971a,b) characterize efficient production and tax rules when the government must raise revenue with distortionary instruments. The long-run capital-tax result in Chamley (1986) shows why a tax on the return to an accumulated factor can have a large stationary cost. The present model is narrower than those general results. It compares three proportional tax bases in a quantitative two-sector economy and introduces collection costs directly in the resource constraint.

The second body of work studies taxation when part of economic activity is informal. Emran and Stiglitz (2005) show that incomplete value-added-tax coverage changes standard indirect-tax prescriptions. Gordon and Li (2009) explain several features of developing-country tax systems through firms' ability to avoid taxes outside the financial system. Keen (2007) emphasizes that a value-added tax can still reach informal operators through taxed formal inputs and imports. This paper captures that partial reach with a parameter that applies the consumption tax to a fraction of informal expenditure.

Informality is an equilibrium allocation, not a fixed untaxed base. La Porta and Shleifer (2014) review its relationship with development. Ulyssea (2018) models firms' extensive and intensive formalization decisions. For Colombia, Osorio-Copete (2016) and Granda and Morales (2023) study how tax policy and labor-market rigidities affect informality. The present paper uses a smooth labor-regime choice. The purpose is not to reproduce every margin of firm informality, but to preserve the feedback from a formal-payroll tax to the number of formal workers and therefore to its own revenue base.

The third body of work concerns public debt. In the tax-smoothing argument of Barro (1979), debt can distribute temporary fiscal pressures across time when the excess burden of taxation is convex. Bohn (1998) studies the positive response of primary surpluses to debt as a condition for fiscal sustainability. The fiscal rule used below combines both ideas: debt evolves through the government budget, and the primary surplus responds positively when debt lies above its anchor. The aging shock is permanent, however. Debt can smooth its arrival but cannot eliminate its present-value tax cost.

Finally, the Colombian pension literature documents the size and fiscal risks of the existing system (Ospina-Tejeiro et al., 2024). This paper does not model a specific pension regime. It studies the financing problem created by any old-age outlay that remains fixed per beneficiary while the number of beneficiaries rises relative to the working-age population.

3 Model

3.1 Population aging and social-security spending

The economy is normalized by the working-age population. The old-age dependency ratio is $d_t = N_t^o/N_t^y$. A fixed real outlay $\bar{g}_o$ per older adult implies

$$G_t^{SS} = \bar{g}_o d_t. \tag{1}$$

Social-security outlays combine transfers and real public services. Let $\zeta \in [0,1]$ denote the share that purchases health, care, administration, or other real resources:

$$G_t^R = \zeta G_t^{SS}, \qquad T_t^o = (1-\zeta)G_t^{SS}. \tag{2}$$

Transfers redistribute purchasing power inside the representative dynasty. Only G_t^R absorbs goods in the aggregate resource constraint. This distinction prevents a pension payment from being treated mechanically as if the transferred resources disappeared.

3.2 Household, consumption, and asset accumulation

The representative dynasty maximizes

$$\sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}-1}{1-\sigma} + \chi_G \log G_t^R \right]. \tag{3}$$

The benchmark sets $\chi_G = 0$. It therefore asks which tax mix preserves private consumption, not whether additional social-security services produce benefits.

Formal and informal consumption are imperfect substitutes:

$$C_t = \left[\varpi^{1/\eta}(C_t^F)^{(\eta-1)/\eta} + (1-\varpi)^{1/\eta}(C_t^I)^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)}, \tag{4}$$

where η is the elasticity of substitution and ϖ is the formal-good weight. The formal producer price is normalized to one. The consumer prices are

$$\tilde{p}_t^F = 1+\tau_t^c, \qquad \tilde{p}_t^I = p_t^I(1+\xi_I \tau_t^c), \tag{5}$$

where $\xi_I \in [0,1]$ is the fraction of informal expenditure reached indirectly by the consumption tax. The CES price index and relative demand are

$$P_t^C = \left[\varpi(\tilde{p}_t^F)^{1-\eta} + (1-\varpi)(\tilde{p}_t^I)^{1-\eta}\right]^{1/(1-\eta)}, \tag{6}$$

$$\frac{C_t^F}{C_t^I} = \frac{\varpi}{1-\varpi}\left(\frac{\tilde{p}_t^F}{\tilde{p}_t^I}\right)^{-\eta}. \tag{7}$$

The household holds productive capital and government bonds. Its consolidated budget can be written as

$$\begin{aligned} P_t^C C_t + K_{t+1} + B_{t+1} = (1-\tau_t^p) w_t^F L_t^F + w_t^I L_t^I \\ + \left[1-\delta+(1-\tau_t^k) r_t^K\right] K_t + (1+r_t^b) B_t + T_t^o. \end{aligned} \tag{8}$$

The steady-state Euler conditions for two assets held in positive amounts are

$$1 = \beta\left[1-\delta+(1-\tau^k) r^K\right] = \beta(1+r^b). \tag{9}$$

The calibration sets $\beta = 1/(1+r^b)$, so the stationary bond and capital returns are consistent.

3.3 Formal and informal production

Formal firms use capital and formal labor:

$$Y_t^F = A_F K_t^\alpha (L_t^F)^{1-\alpha}. \tag{10}$$

Competitive factor prices are

$$w_t^F = (1-\alpha)\frac{Y_t^F}{L_t^F}, \qquad r_t^K = \alpha\frac{Y_t^F}{K_t}. \tag{11}$$

Informal production is labor intensive:

$$Y_t^I = A_I L_t^I, \qquad w_t^I = p_t^I A_I. \tag{12}$$

Labor satisfies $L_t^F + L_t^I = 1$. The assumption that only formal production uses capital gives payroll and capital taxation a common formal-sector base, while preserving different intensive margins.

3.4 Endogenous labor informality

Workers receive idiosyncratic preferences for the two labor regimes. With a logistic difference in those preferences, the formal share is

$$L_t^F = \Lambda\left(\frac{\log[(1-\tau_t^p)w_t^F] - \log(w_t^I) - \kappa}{\mu}\right), \qquad \Lambda(x) = \frac{1}{1+e^{-x}}. \tag{13}$$

The fixed cost κ shifts the average return to formal employment and μ controls the dispersion of regime preferences. For finite μ, both labor regimes coexist. A payroll tax lowers the net formal wage and increases informality. A consumption tax affects informality indirectly through the relative demand for formal and informal goods.

3.5 Government revenue, debt, and the fiscal rule

The government raises revenue from consumption, formal payroll, and capital income:

$$R_t^c = \tau_t^c\left(C_t^F + \xi_I p_t^I C_t^I\right), \tag{14}$$

$$R_t^p = \tau_t^p w_t^F L_t^F, \tag{15}$$

$$R_t^k = \tau_t^k r_t^K K_t, \tag{16}$$

$$R_t = R_t^c + R_t^p + R_t^k. \tag{17}$$

Debt follows

$$B_{t+1} = (1+r_t^b)B_t + G_t^{SS} - R_t. \tag{18}$$

Let $q_t = B_t/Y_t^N$, where nominal output is $Y_t^N = Y_t^F + p_t^I Y_t^I$. If trend output grows at rate γ, the primary surplus ratio that stabilizes debt at q is $(r^b-\gamma)q$. The fiscal reaction function is

$$\frac{R_t - G_t^{SS}}{Y_t^N} = (r_t^b - \gamma)q_t + \phi_q(q_t - \bar{q}). \tag{19}$$

The feedback coefficient $\phi_q > 0$ raises the surplus when debt exceeds the anchor. In the stationary experiments $q = \bar{q}$, so the second term is zero.

3.6 Collection costs and market clearing

Collecting a higher rate from a given base uses real resources. The reduced form is

$$\Psi_t = \frac{1}{2}\sum_{j\in\{c,p,k\}} \chi_j(\tau_t^j)^2 \mathcal{B}_t^j, \tag{20}$$

where $\mathcal{B}_t^j$ is the corresponding base in equations (14)–(16). The term includes tax administration, private compliance, enforcement, and avoidance resources. It does not represent revenue lost to the government; it is a real cost in the formal-goods market.

Capital accumulation and goods-market clearing are

$$K_{t+1} = (1-\delta)K_t + I_t, \tag{21}$$

$$Y_t^F = C_t^F + I_t + G_t^R + \Psi_t, \tag{22}$$

$$Y_t^I = C_t^I. \tag{23}$$

At a stationary equilibrium, $I = \delta K$. Pension transfers do not enter equation (22); their recipients use them to purchase the goods already included in private consumption.

3.7 Tax mix and policy problem

Let $\boldsymbol{\omega} = (\omega_c, \omega_p, \omega_k)$ denote the shares of required revenue assigned to the three bases:

$$R^j = \omega_j R, \qquad \omega_c + \omega_p + \omega_k = 1, \qquad \omega_j \geq 0. \tag{24}$$

Given an equilibrium allocation, tax rates satisfy

$$\tau^j = \frac{\omega_j R}{\mathcal{B}^j}. \tag{25}$$

For each dependency ratio and debt anchor, the stationary Ramsey problem is

$$\max_{\boldsymbol{\omega} \in \Delta^2} \frac{u(C(\boldsymbol{\omega}))}{1-\beta} \quad \text{subject to equations (1)–(25).} \tag{26}$$

Because $u(C)$ is increasing and $\chi_G = 0$, this is equivalent to maximizing stationary composite consumption. The tax mix is chosen on a simplex; the model does not impose an interior solution.

4 Calibration and numerical experiment

Table 1 separates the two observed targets, the parameters chosen to match them, and the remaining benchmark inputs. Formal production uses a capital share of 0.33 and annual depreciation of 0.06. A four-percent real bond rate implies $\beta = 1/1.04$. The CES elasticity is 1.5: formal and informal goods are substitutes, but far from perfect substitutes. The consumption tax reaches 15 percent of informal expenditure.

Table 1: Benchmark calibration

Quantity	Symbol	Value	Unit or source
Observed calibration targets			
Public pension spending	G^S/Y	0.057	Share of GDP
Labor informality	$1-n_F$	0.554	Share of employment
Parameters calibrated to the observed targets			
Outlay per older adult	$\bar{g}_o$	0.426	Model consumption units
Formal-work fixed cost	κ	0.633	Model utility units
Benchmark inputs and provisional assumptions			
Formal capital share	α	0.330	Provisional
Annual depreciation	δ	0.060	Provisional
Annual discount factor	β	0.962	Implied by r^b
CES elasticity	η	1.500	Provisional
Informal VAT coverage	ξ_I	0.150	Provisional
Regime-choice scale	μ	0.300	Provisional
Real-resource share	ζ	0.250	Provisional
Real bond rate	r^b	0.040	Provisional
Trend output growth	γ	0.010	Provisional
Debt anchor	$\bar{q}$	0.550	Law 2155 of 2021
Common collection cost	χ_j	2.000	Provisional

Notes: The model jointly chooses $\bar{g}_o$ and κ to match the two observed targets. The value $\bar{g}_o = 0.426$ is expressed in model consumption units per older adult; it is not a share of GDP or a tax rate. This calibration is performed in an environment that assigns 45 percent of required revenue to consumption taxes, 35 percent to payroll taxes, and 20 percent to capital-income taxes. That mix is neither an observed target nor an estimate or policy result. Effective tax rates and quantities in model units should not be read as statutory Colombian rates or quantities in pesos.

Two parameters match Colombian targets under the reference revenue mix. $\bar{g}_o$ matches public pension spending of 5.7% of GDP (Organisation for Economic Co-operation and Development, 2023). κ matches 55.4% national labor informality (Departamento Administrativo Nacional de Estadística, 2025a). The reference mix is a normalization, not an estimate of the composition of Colombian tax revenue.

The debt anchor is 55 percent of GDP. Law 2155 defines that value for net debt of the central government and a 71-percent limit (República de Colombia, 2021). The counterfactuals use anchors of 40, 55, and 70 percent. They keep the structural parameters and $\bar{g}_o$ fixed.

The demographic experiment reads every annual value between 2024 and 2070 from DANE's national projections (Departamento Administrativo Nacional de Estadística, 2025b). For each year, the code solves a stationary equilibrium conditional on that year's d_t. The resulting annual line is a sequence of conditional steady states. It is not a perfect-foresight transition with predetermined capital and debt.

The collection-cost coefficients are equal across instruments. This choice does not favor one base by assigning it a lower primitive collection cost. Their common level is not empirically estimated. I therefore report the frictionless case $\chi_j = 0$ to identify which conclusion depends on convex collection costs.

5 Results

5.1 The stationary tax mix

Table 2 reports the tax mix that maximizes stationary composite consumption at three points on the DANE demographic path. All shares and rates are percentages. In 2024, the model assigns 92.5% of required revenue to consumption and 7.5% to formal payroll. In 2050, the shares are 82.5% and 17.5%. By 2070, the payroll share reaches 25.0%.

Table 2: Stationary optimal tax mix along the demographic path

Year	ω_c	ω_p	ω_k	τ_c	τ_p	Informality
2024	92.5	7.5	0.0	11.3	1.1	54.7
2050	82.5	17.5	0.0	17.4	4.5	56.0
2070	75.0	25.0	0.0	30.4	11.1	58.0

Notes: ω_j is the share of required revenue assigned to tax j; τ_j is the corresponding effective rate. Every entry other than the year is a percentage. The debt anchor is 55 percent of GDP.

The capital-income share is zero in all three stationary optima. The result has a precise model interpretation. Capital is used only in formal production, and taxing its return lowers the stationary capital stock through the Euler condition. The representative household eliminates any redistributive reason to tax capital, and formal firms earn no rents. The zero result is therefore consistent with this specification. It is not a general recommendation to eliminate capital taxation in Colombia.

The payroll share rises as aging increases because collection costs are convex. Concentrating an ever larger revenue requirement in consumption raises the marginal resource cost of that instrument. A moderate payroll tax then costs less than the additional collection resources required by a still higher consumption rate, despite the payroll tax's direct effect on informality.

Figure 1 shows the 2050 policy search. Each point is a feasible revenue mix. The lower-left vertex assigns all revenue to consumption, the lower-right vertex assigns all revenue to payroll, and the top vertex assigns all revenue to capital income. Darker points have larger consumption-equivalent losses, capped at eight percent for the color scale. The star marks the optimum at 82.5 percent consumption and 17.5 percent payroll. Moving toward the capital vertex lowers consumption sharply.

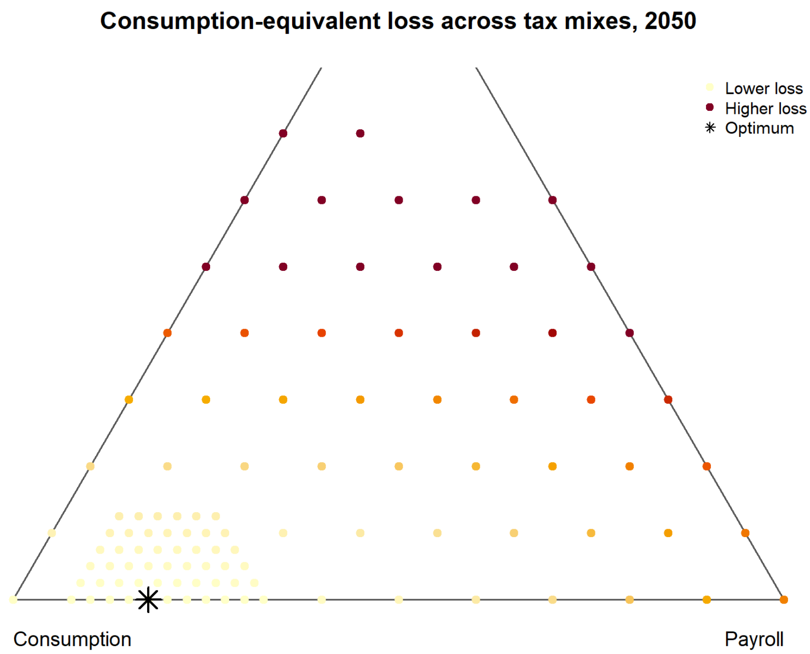

Figure 1: Consumption-equivalent loss across tax mixes in 2050

Notes: The triangle contains the feasible coarse grid and the local refinement. Losses are measured relative to the best stationary composite consumption in the grid. The color scale is capped at eight percent.

Figure 2 separates revenue shares from rates. The left panel shows the shift from consumption toward payroll as d_t rises. The right panel shows that the effective consumption rate rises from 11.3 percent in 2024 to 17.4 percent in 2050 and 30.4 percent in 2070. The payroll rate rises from 1.1 to 4.5 and 11.1 percent. The 2070 rates are not a forecast. They show the pressure created by holding the outlay per older adult fixed while the dependency ratio more than triples relative to its 2024 level.

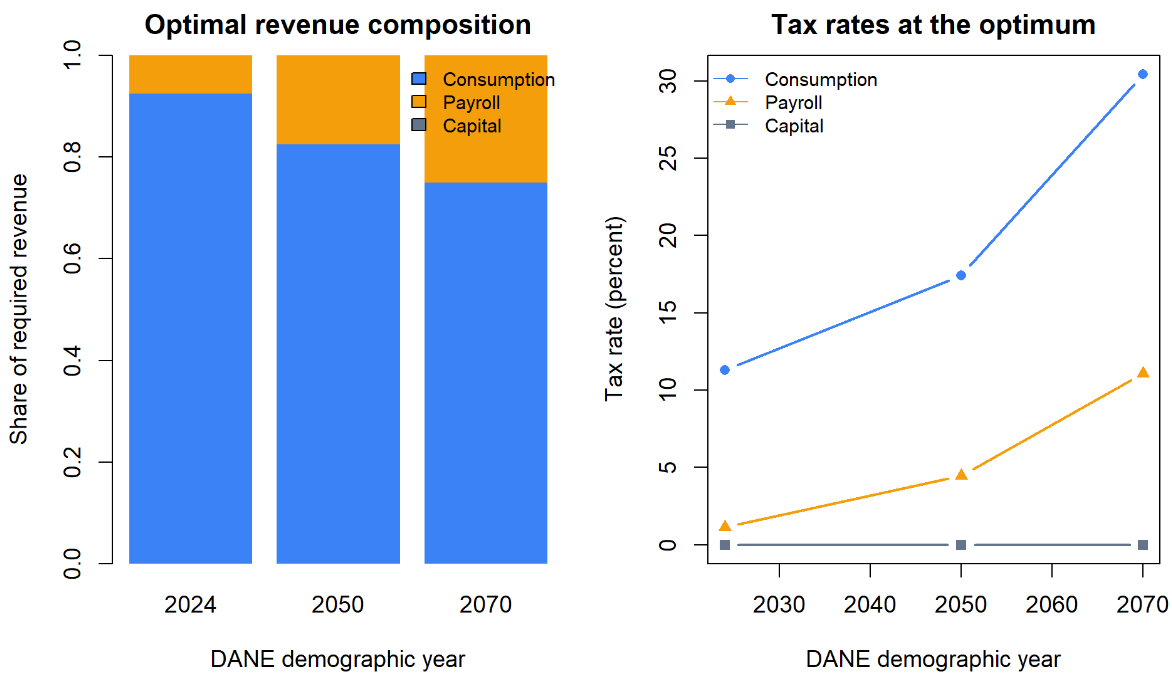

Figure 2: Optimal revenue composition and effective tax rates

5.2 What generates an interior mix?

The imperfect substitution between formal and informal goods changes the incidence of the consumption tax. A higher τ^c raises the relative price of formal consumption and shifts expenditure toward the informal good. That mechanism erodes the formal base and changes labor demand. It does not, however, produce an interior optimum in this calibration.

When all three collection-cost coefficients are set to zero, the 2050 optimum assigns 100.0% of revenue to consumption. Payroll and capital taxes are both zero. This comparison identifies the source of the benchmark mix: the CES demand system creates an informal-consumption response, but convex collection costs create the gain from spreading revenue across bases.

This result also identifies the paper's main calibration requirement. The level and instrument-specific values of χ_j must be estimated before the numerical shares can support a policy recommendation. The current common coefficient provides a transparent benchmark, not empirical identification.

5.3 Conditional demographic paths

Figure 3 compares the reference revenue mix with the 2024-optimal mix held fixed through 2070. The upper-left panel reports social-security outlays and required revenue under the reference mix. The upper-right panel reports the three effective rates under the fixed 2024-optimal mix. The lower panels compare informality and the stationary capital stock.

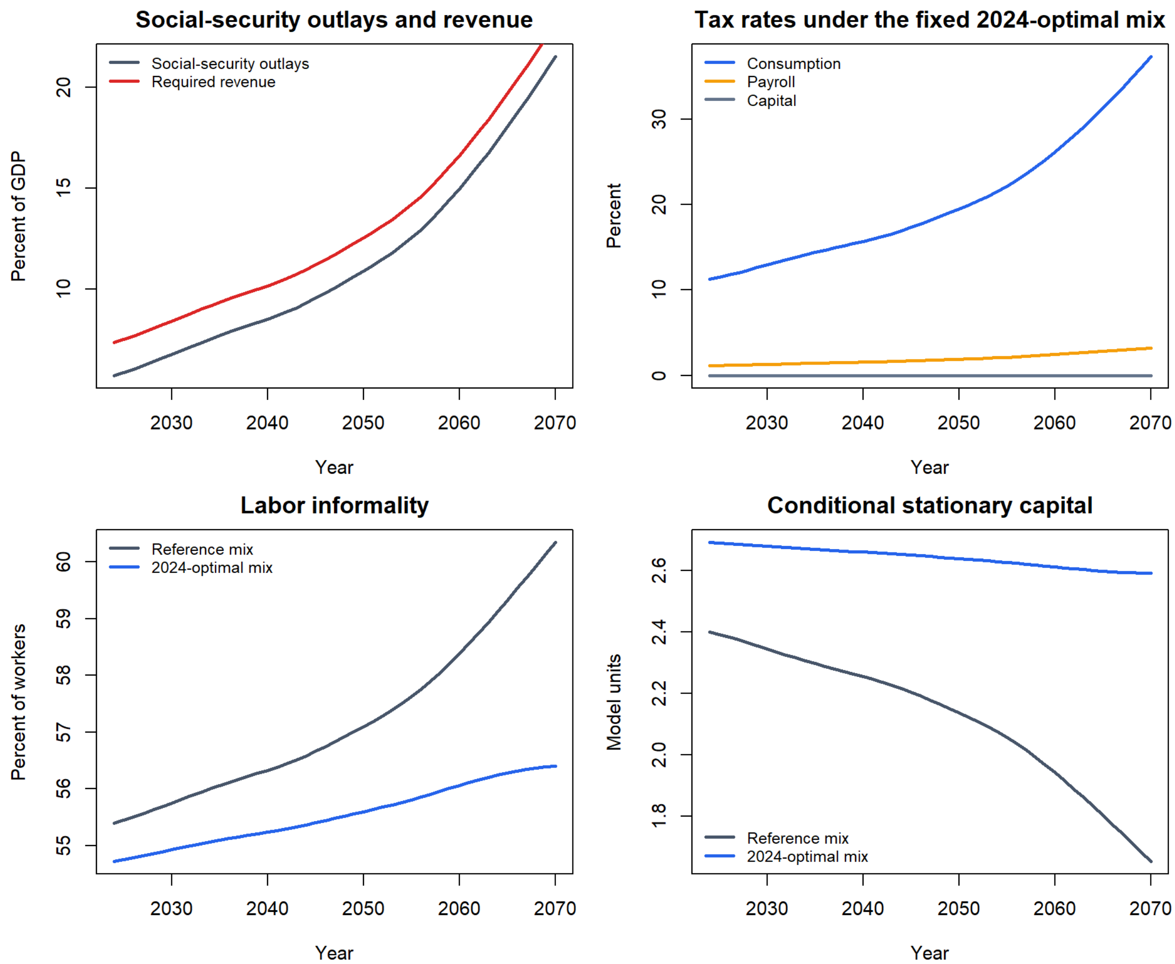

Figure 3: Conditional stationary equilibria along the DANE demographic path

Notes: The reference revenue mix is 45 percent consumption, 35 percent payroll, and 20 percent capital income. The alternative holds the 2024-optimal shares fixed at 92.5, 7.5, and zero percent. Each annual observation is a stationary equilibrium conditional on that year's DANE dependency ratio.

Under the reference mix, informality rises from 55.4 percent in 2024 to 57.1 percent in 2050 and 60.3 percent in 2070. The stationary capital stock falls from 2.399 to 2.137

and 1.654 model units. Under the fixed 2024-optimal mix, informality reaches 55.6 percent in 2050 and 56.4 percent in 2070, while capital remains above 2.59. Composite consumption in 2070 is 0.835 under the fixed 2024-optimal mix and 0.792 under the reference mix.

The figure does not imply that capital jumps to its next stationary value each year. It reports how the long-run allocation associated with each demographic ratio changes. A true transition would impose the predetermined 2024 capital stock, solve households' intertemporal choices, and let the fiscal rule determine debt and taxes along one perfect-foresight path.

5.4 The debt anchor

Figure 4 varies $\bar{q}$ in the 2050 demographic equilibrium and reoptimizes the tax mix. Raising the anchor from 40 to 70 percent of GDP raises required revenue from 11.18 to 12.13 percent of GDP. Composite consumption falls from 0.924 to 0.918, and informality rises from 55.8 to 56.1 percent. The statutory 55-percent anchor lies between those outcomes.

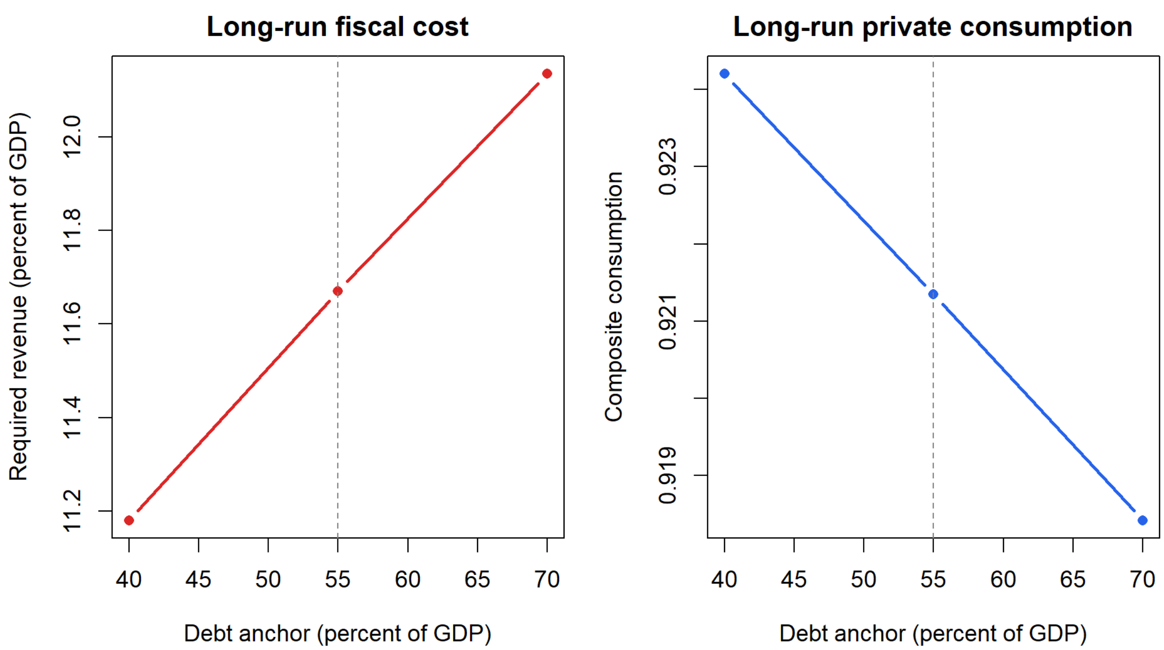

Figure 4: Debt-anchor sensitivity in the 2050 demographic equilibrium
Notes: The dashed line marks Colombia's statutory 55-percent net-debt anchor. The tax mix is reoptimized at each anchor. The exercise compares stationary equilibria and does not measure transitional tax smoothing.

A larger $\bar{q}$ is not a free financing source. At a stationary debt ratio, the government must raise a primary surplus of $(r^b - \gamma)\bar{q}$ to service the debt net of trend growth. A higher anchor therefore raises long-run taxes and collection costs. Debt can still be useful during the transition if it prevents tax rates from changing abruptly, as in the tax-smoothing logic of Barro (1979). Quantifying that benefit requires a dynamic transition and cannot be inferred from Figure 4.

6 Limitations and extensions

The numerical ranking is conditional on six restrictions.

First, the model has one representative dynasty. It contains no intragenerational variance of consumption and no difference between young and old households' marginal utilities. The welfare criterion therefore cannot measure the regressivity of consumption taxes or the distribution of benefits and tax burdens across generations. Adding household types is necessary before the paper can answer a distributional version of the research question.

Second, $\zeta = 0.25$ is provisional. Pension payments are transfers, whereas health and long-term care absorb labor and goods. The aggregate effect of aging depends on this composition. Administrative records should separately calibrate cash transfers and real services per older adult.

Third, the collection-cost coefficient is not estimated. The frictionless comparison shows that it changes the qualitative result from a corner to a mix. Empirical evidence on compliance, enforcement, and base erosion by instrument is therefore more important for the tax shares than another digit of numerical precision in the current grid.

Fourth, the tax rates are reduced-form effective rates. The consumption tax combines statutory rates, exemptions, and incomplete informal coverage. The payroll tax combines mandatory contributions and other formal-employment costs. The capital tax does not distinguish normal returns from rents. Direct comparison with Colombian statutory rates would be misleading.

Fifth, the annual paths are conditional stationary equilibria. They do not contain capital and debt transition dynamics, unexpected demographic news, or cohort-specific saving. The next version should solve the perfect-foresight path with K_{2024} and B_{2024} predetermined.

Sixth, $\bar{q}$ is a policy parameter rather than an optimized state. An endogenous debt anchor requires a sovereign-risk premium, an explicit probability of fiscal stress, or an external-debt margin. Without one of those costs, the transition benefit of borrowing is not identified separately from the stationary interest bill.

7 Conclusions

Population aging converts a fixed real outlay per older adult into a rising revenue requirement per working-age adult. DANE's projected dependency ratio almost doubles between 2024 and 2050 and exceeds 48 percent by 2070. The government's financing choice therefore changes the long-run allocation even when the benefit received by each older adult remains constant.

In the benchmark model, the stationary tax mix relies primarily on consumption and secondarily on formal payroll. The payroll share rises as the population ages because convex collection costs make it increasingly costly to concentrate the entire revenue requirement in consumption. The capital-income share is zero because capital accumulation is elastic and the representative household removes distributional motives for capital taxation.

Two negative results are equally important. Imperfect substitution between formal and informal consumption does not by itself prevent a corner solution: without collection costs, the model finances everything through consumption. Public debt also does not remove the permanent tax requirement. A higher debt anchor raises stationary interest payments and lowers consumption.

The current results establish a mechanism, not a Colombian tax package. The next empirical step is to estimate the collection and base-erosion costs of each instrument. The next modeling step is to add heterogeneous young and old households and solve the full transition. Those additions would allow the paper to compare efficiency, intragenerational incidence, and intergenerational welfare within the same policy experiment.

A Numerical algorithm

The replication uses base R and proceeds in six steps.

1. Demographic input. The script reads annual DANE populations aged 15–64 and 65 and over for 2024–2070. It constructs $d_t = N_t^o / N_t^y$ without interpolation.

2. Calibration. At the 2024 dependency ratio, the code chooses $(\bar{g}_o, \kappa)$ to minimize the squared distance from the 5.7-percent spending target and 55.4-percent informality target under the reference revenue mix. The outer problem uses Nelder–Mead.

3. Stationary equilibrium. For a given $(d, \boldsymbol{\omega}, \bar{q})$, the unknown vector is

$$z = \left(\log K, \operatorname{logit} L^F, \log p^I, \log C^F\right).$$

The transformations impose $K > 0$, $p^I > 0$, $C^F > 0$, and $L^F \in (0, 1)$. From z, the code computes production, factor prices, social-security spending, required revenue, tax bases, tax rates, and collection costs.

The four residuals are the formal-labor choice (13), CES relative demand (7), capital Euler condition (9), and formal-goods market (22). BFGS minimizes the sum of squared residuals. Four dispersed starting values are used for a standalone solution. Adjacent tax mixes use the preceding equilibrium as a warm start. Nelder–Mead is a fallback when the residual norm remains above tolerance.

4. Policy search. The code first evaluates a simplex grid with increments of 0.10. It then evaluates a local grid with increments of 0.025 around the best coarse point. The reported optimum is the feasible point with the highest stationary utility. The procedure does not impose positive shares.

5. Counterfactuals. The tax mix is optimized at the 2024, 2050, and 2070 dependency ratios. The 2050 problem is repeated for debt anchors of 40, 55, and 70 percent. The frictionless counterfactual sets every χ_j to zero. Annual conditional paths hold either the reference mix or the 2024-optimal mix fixed.

6. Verification. Automated tests require the calibrated equilibrium to match both targets, each equilibrium residual to be below tolerance, tax rates to be nonnegative, the policy grid to contain one optimum, and social-security spending as a share of output to rise across selected demographic years. All figures and LaTeX macros are generated from the same result files used in the text.

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